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Wireshark Network Traffic Analysis Report

1.Objective

This report analyzes real network traffic captured using Wireshark focusing on DNS resolution and encrypted HTTPS communication.

The goal is to understand how client-server communication occurs, how DNS queries are resolved and how encrypted HTTPS packets appear in packet captures.

2. Tools used

- Wireshark
- macOS
- TCP/IP stack
- Network interface: en0 (Wi-Fi)

3.Methodology

- Captured live network traffic on the local Wi-Fi interface.
- Applied protocol filters:
  - DNS: dns
  - HTTPS: tcp.port == 443
- Selected individual packets to inspect:
  - Source/ Destination IP

- Ports
- DNS query/ response structure
- Protocol layers (Ethernet, IP, TCP)
- TLS encrypted payloads

- Documented findings with screenshots.

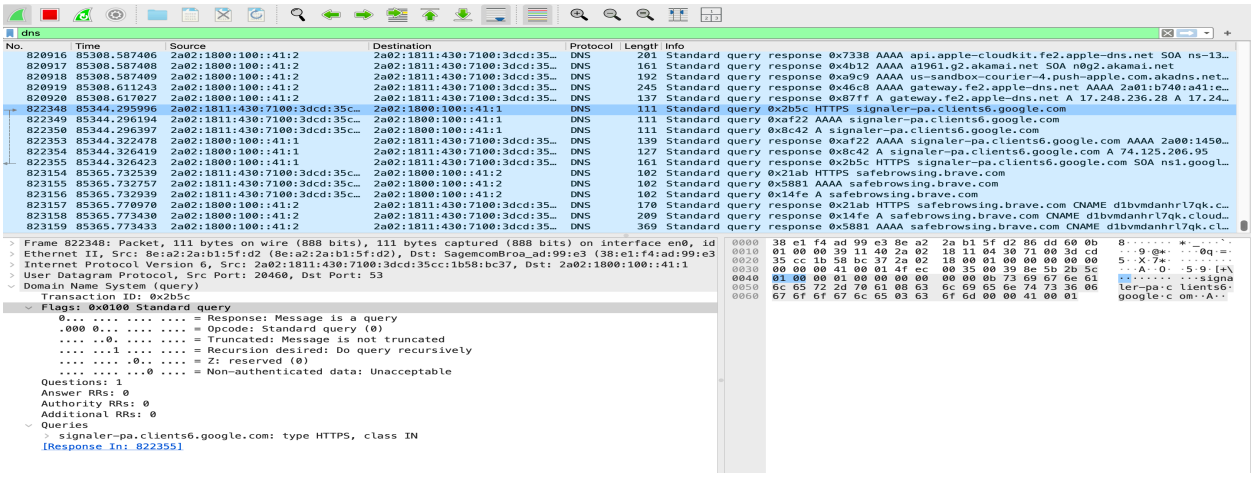
4.Findings

4.1 DNS Traffic Analysis

DNS queries and responses were successfully captured.

The captured packets show:

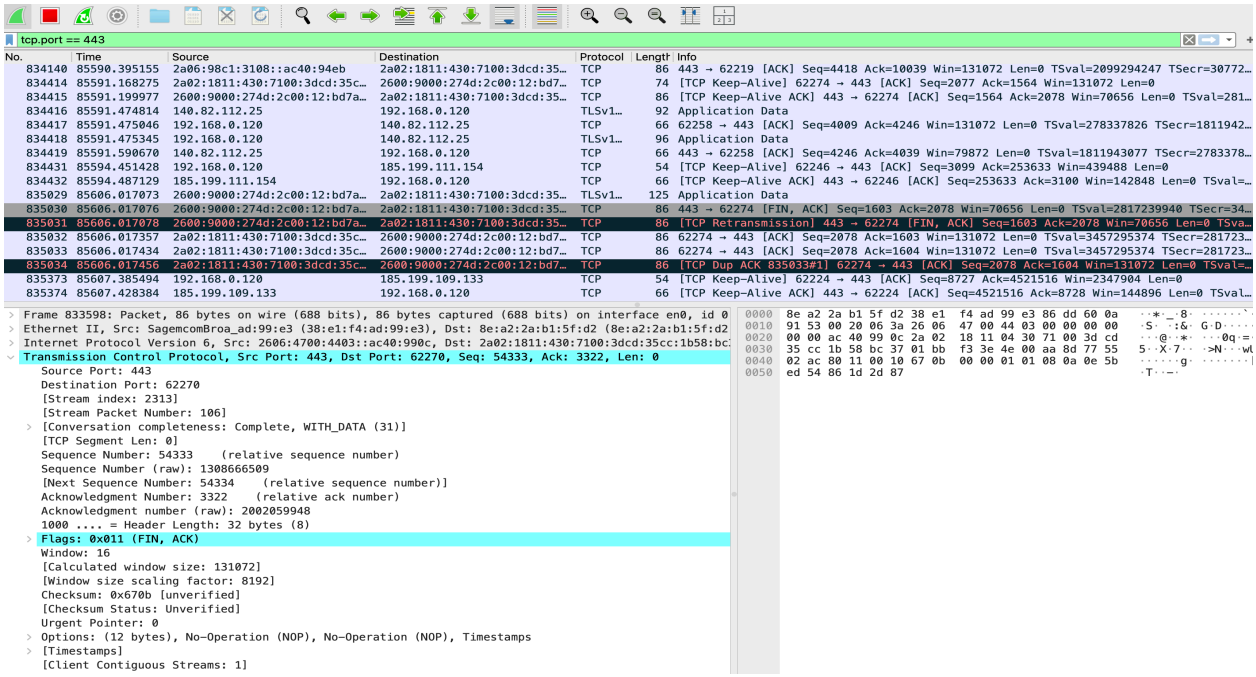
- AAAA queries for domains such as
  - lcdrn-locator.apple.com
  - imap.gmail.com
  - safebrowsing.brave.com
- Matching Transaction IDs between queries and responses
- IPv6-based DNS resolution
- Additional RRs and authority sections depending on the domain.



4.2 HTTPS Traffic Analysis

HTTPS traffic was isolated using the filter tcp.port == 443.

- Key observations:
  - Traffic is encrypted (TLS Application Data)
  - No readable content is visible (unlike HTTP)
  - TCP Keep-Alive ACK packets maintain the session
  - Source Port: 443 (server)
  - Destination port: ephemeral client ports (e.g 62219, 62720)
  - Flags such as PSH, ACK indicate active data transfer



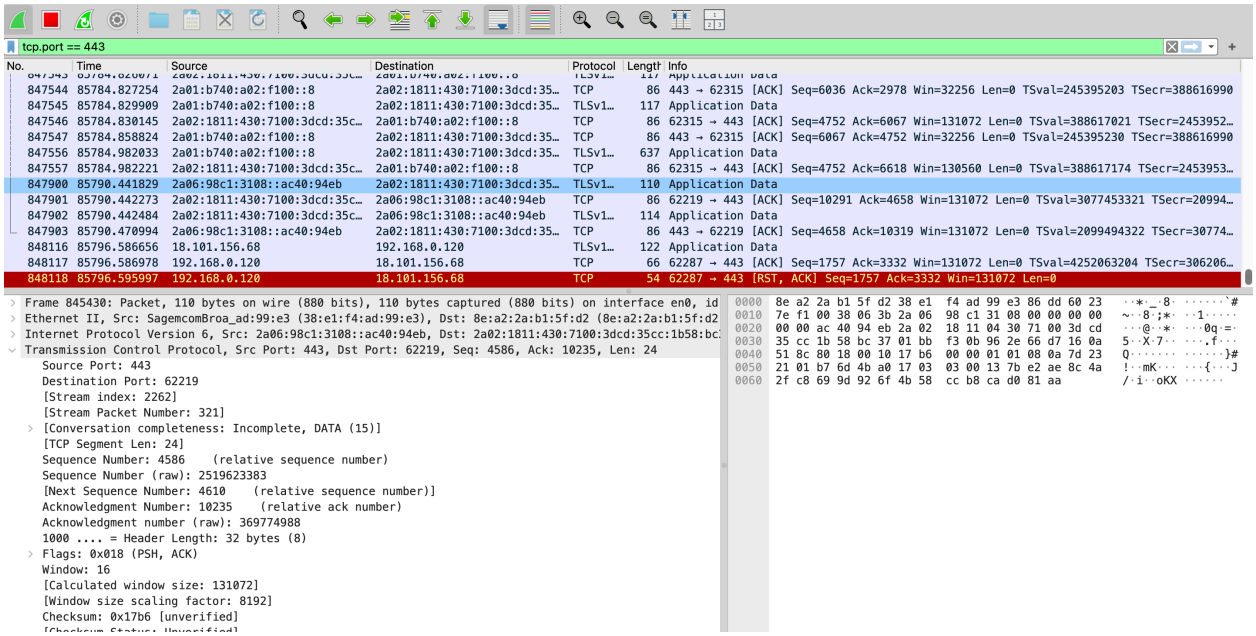
4.3 Packet Flow (Detailed Packet Structure)

A selected HTTPS packet shows the full protocol stack:

- Ethernet 2: MAC addresses
- IPv6: Source and destination addresses

- TCP:
    - Source Port: 443
    - Flags: PSH, ACK
    - Sequence and acknowledgement numbers
    - Window size and checksum
- Encrypted TLS payload visible in Packet Bytes

This demonstrates how encrypted HTTPS traffic appears in Wireshark.



5. Conclusion

The Wireshark analysis successfully captured and examined both DNS and HTTPS traffic.

DNS packets revealed readable domain resolution behaviour while HTTPS packets showed fully encrypted TLS payloads, demonstrating how secure communication appears in packet captures.

By inspecting Packet Details and Packet Bytes, I gained a deeper understanding of how data flows through Ethernet, IP, TCP and TLS layers.
This exercise strengthened my foundational skills in network analysis, encryption concepts and protocol behaviour.

- Key takeaways:
  - DNS traffic is readable and shows how domain names are resolved.
  - HTTPS traffic is encrypted, showing only TCP/TLS metadata.
  - Packet Details and Packet Bytes views help visualize how data moves through each protocol layer.
  - Understanding encrypted vs unencrypted traffic is essential for cybersecurity and network troubleshooting.